



Digital Communication IC Design

Lecture-03: Channel Model

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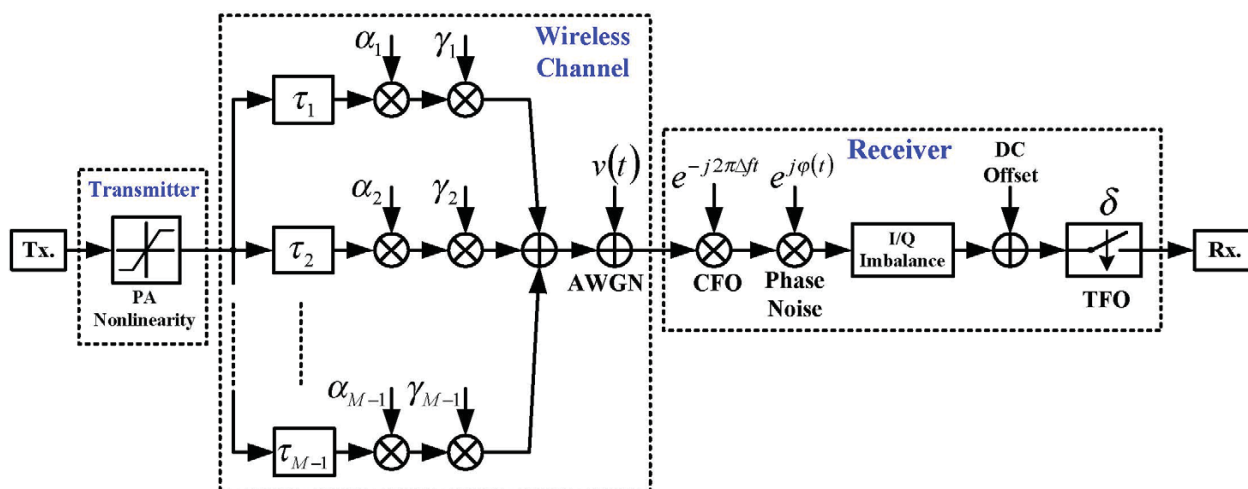


Outline

- ☐ Wireless Channel
- ☐ Wireline Channel

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Wireless Baseband Channel Model



□ Channel model

- Tx. nonlinearity, physical channel (w/ AWGN) and Rx. nonlinearity

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Channel Impairments

□ Three different propagation mechanisms

- Path loss
- Shadowing
- Multipath fading

□ Path loss

- Radio signals propagating through free space at a rate that is inversely proportional to the squared distance between the Tx. and the Rx.

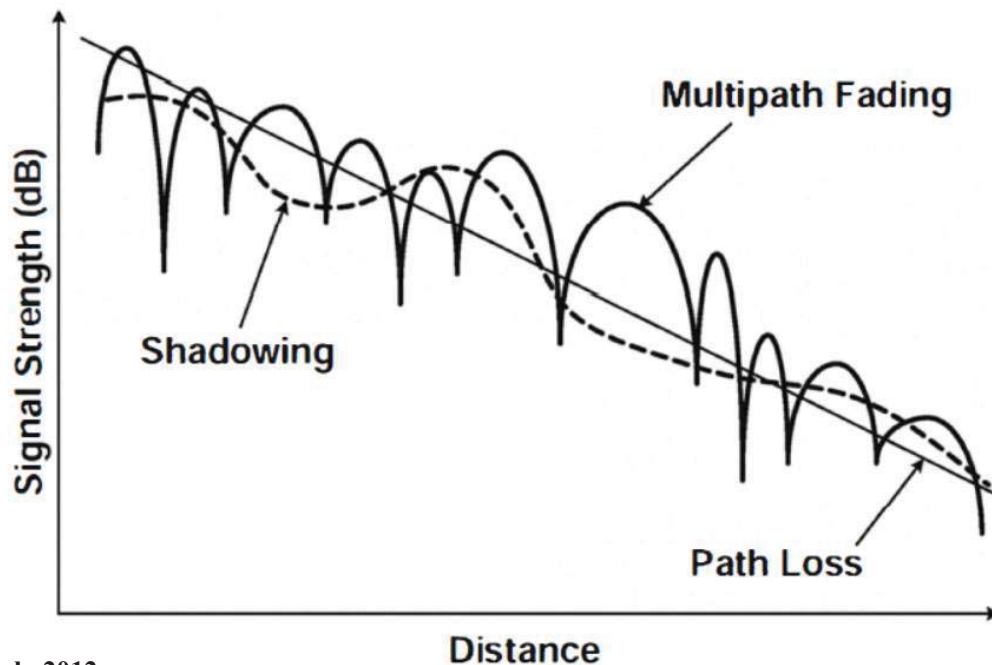
$$P_r(d) \propto \left(\frac{\lambda}{4\pi d} \right)^2$$

- $P_r(d)$: the received power at distance d from the Tx.
- λ : the wavelength of the carrier signal

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Channel Model with Path Loss, Shadowing and Multipath Fading



T. Z. Chiueh, 2012

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Path Loss

- ❑ Line of sight (LOS) and non-LOS (NLOS)
 - LOS: no obstruction between Tx. and Rx.
 - NLOS: It exists in the most wireless communication environments.

❑ A simplified log-distance path loss model

- Represent the degree of signal power attenuation as the distance between the Tx. And Rx. increasing

$$L = 10n \log \left(\frac{d}{d_{ref}} \right) + L_{ref} \rightarrow PL(d) = PL(d_{ref}) + 10n \log \left(\frac{d}{d_{ref}} \right)$$

- n : the path loss exponent
- d : the separation between Tx. and Rx.
- L_{ref} : the path loss value in free space for the reference distance d_{ref}

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Path Loss (cont'd)

- The path loss with the shadowing effect

$$PL(d) = PL(d_{ref}) + 10n \log\left(\frac{d}{d_{ref}}\right) + X$$

- X : the shadowing effect in dB scale with zero-mean Gaussian random variable
- Give rise to the signal strength variation of 20 dB
- The standard deviation of the shadowing variable ≥ 10 dB (in some cellular and indoor environments)

Environment	Path Loss Exponent, n
Free space	2
Urban area cellular radio	2.7 to 3.5
Shadowed urban cellular radio	3 to 5
In building line-of-sight	1.6 to 1.8
Obstructed in building	4 to 6
Obstructed in factories	2 to 3

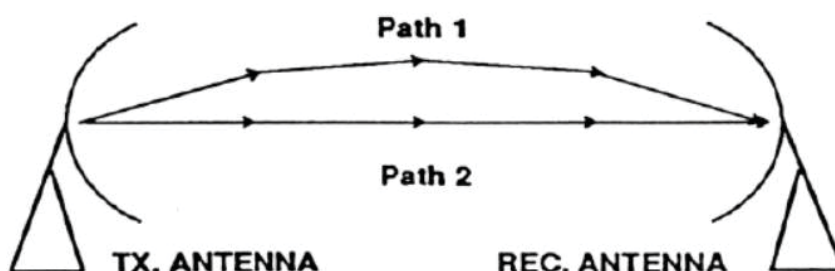
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Multipath Fading

- Two types of multipath
 - Variation between arrival times of different paths (flat/frequency-selective fading)
 - Relative motion speed between the Tx. and the Rx. (slow/fast fading)



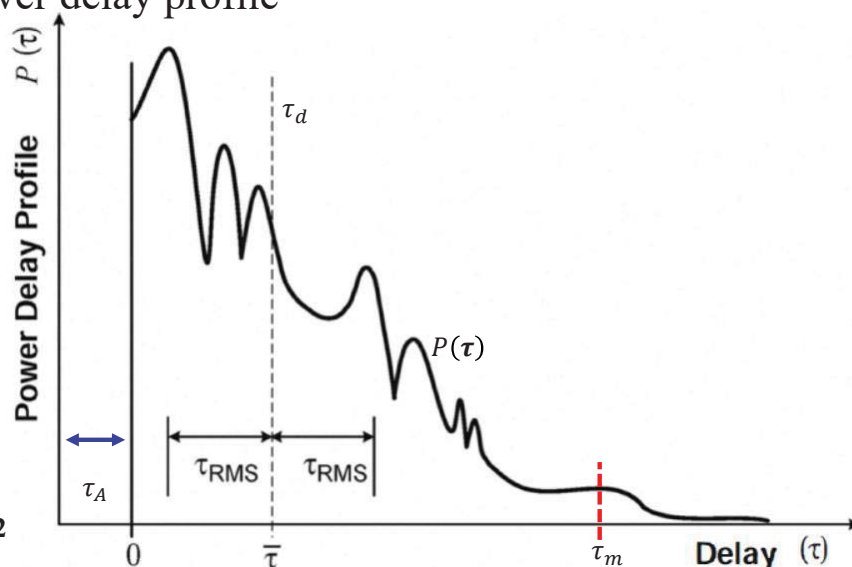
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Multipath Fading (cont'd)

□ Delay spread (dispersion)

- First arrival delay, mean excess delay, RMS delay spread and maximum excess delay
- Power delay profile



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Multipath Fading (cont'd)

□ The baseband complex impulse response of a multipath channel

$$h(\tau; t) = \sum_r \beta_r(t) e^{j\theta_r(t)} \delta(\tau - \tau_r)$$

- r : the path index
- $\beta_r(t)$: the path gain of the r th path
- $\theta_r(t)$: the phase shift of the r th path
- τ_r : the delay spread of the r th path
- δ : the Dirac delta function

□ Power delay profile

$$P(\tau; t) \equiv |h(\tau; t)|^2 = \sum_r \beta_r^2(t) \delta(\tau - \tau_r)$$

Multipath Fading (cont'd)

- ❑ Root mean square (RMS) delay spread
 - Measure of the amount of signal dispersion in time
 - Assume the channel is stationary.

$$\bar{\tau} = \int_{-\infty}^{\infty} \tau P_{norm}(\tau) d\tau$$

$$P_{norm}(\tau) = \frac{P(\tau)}{\int_{-\infty}^{\infty} P(\tau) d\tau}$$

$$\tau_{RMS} = \sqrt{\int_{-\infty}^{\infty} (\tau - \bar{\tau})^2 P_{norm}(\tau) d\tau}$$

Multipath Fading (cont'd)

- ❑ Taking the Fourier transform of the channel PDP yields the auto-correlation of the channel frequency response $\Phi_{HH}(\Delta f)$ with respect to the frequency difference Δf .
 - If the channel PDP is an impulse function, then the auto-correlation function will become unity over all possible Δf .
 - The channel frequency responses at any two different frequencies are identical (100% correlated) or the channel is all-pass.
- ❑ On the other hand, if the delay spread is relatively large, the auto-correlation function falls off more rapidly.
 - Define the coherence BW B_c :
 - A channel as the frequency displacement Δf at which $\Phi_{HH}(\Delta f)$ become less than a threshold



Multipath Fading (cont'd)

- The channel has rather different characteristics at two frequencies with a difference larger than the coherence BW.

□ Coherent bandwidth (B_c)

$$B_c = \frac{1}{K\tau_{RMS}}$$

- K : range from 5~50 for a threshold from 0.5~ 0.9



Multipath Fading (cont'd)

□ According to the coherent BW and the signal BW, a wireless channel can be divided into as

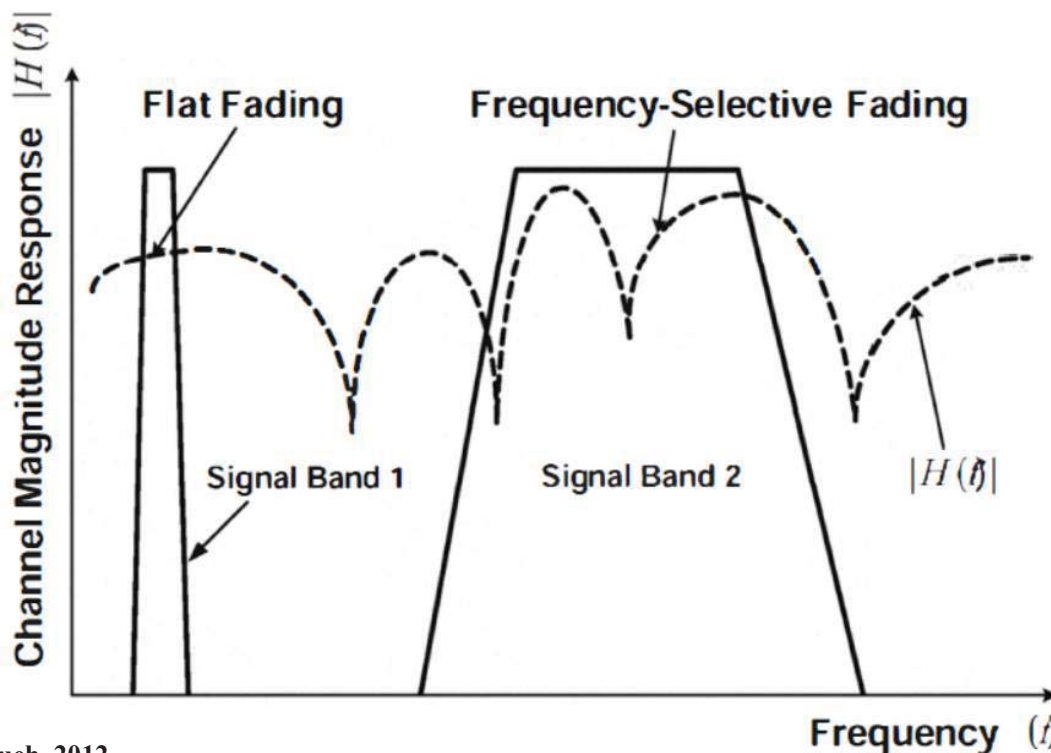
■ Flat fading: Signal BW $\ll$ coherent BW

- The channel frequency response is **flat inside** the signal band.

■ Frequency-selective fading: Signal BW $\ll$ coherent BW

- The different frequency components of the signal suffer **various amplitude attenuations** and **phase shifts**.
- It is hard to equalize the received signal for exact detection.

Coherent BW: Flat and Frequency-Selective Fading



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Types of Small-Scale Fading

Small-Scale Fading

(Based on multipath time delay spread)

Flat Fading

1. BW of signal < BW of channel
2. Delay spread < Symbol period

Frequency Selective Fading

1. BW of signal > BW of channel
2. Delay spread > Symbol period

Small-Scale Fading

(Based on Doppler spread)

Fast Fading

1. High Doppler spread
2. Coherence time < Symbol period
3. Channel variations faster than baseband signal variations

Slow Fading

1. Low Doppler spread
2. Coherence time > Symbol period
3. Channel variations slower than baseband signal variations

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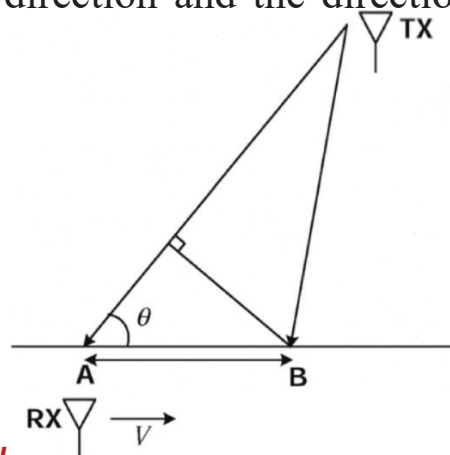
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Doppler Spread – Frequency Dispersion

□ Doppler effect

- Neither the Tx., the receiver, or some scatters is in **motion**, the received signal will be dispersed (frequency shift) in the frequency domain.
- The frequency shift is related to speed of the motion and the angle between the signal arrival direction and the direction of motion.



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Doppler Spread – Frequency Dispersion (cont'd)

□ The received signal with frequency is given as

- $f_{received} = f_c + f_d; f_d = f_c \cdot \frac{v}{c} \cos \theta$

□ Doppler frequency shift f_D and phase shift between A and B

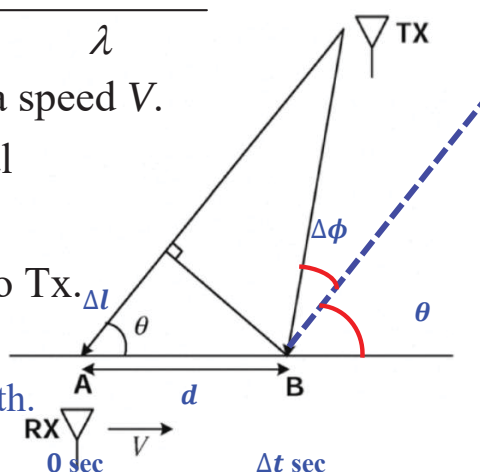
$$f_D = \frac{\Delta \phi}{2\pi \Delta t} = \frac{V \cos \theta}{\lambda} \quad \text{and} \quad \Delta \phi = \frac{2\pi \Delta l}{\lambda} = \frac{2\pi V \Delta t \cos \theta}{\lambda}$$

- The Rx. is moving from A to B with a speed V .
- λ : the wavelength of the carrier signal
- d : the distance between A and B
- Δl : the distance difference from Rx. to Tx.
between A and B

The above consideration only assume one path.

In practice, there are several path between

Tx. and Rx.



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Doppler Spread – Frequency Dispersion (cont'd)

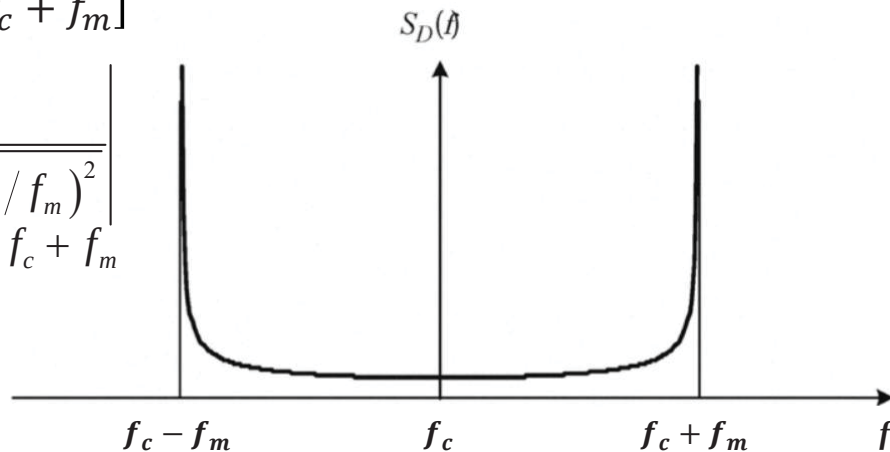
□ Then, the received signal PSD will be spread over a bandwidth limited by the **maximum** Doppler frequency.

■ $f_m = V/\lambda$

■ The classic (Jakes) Doppler spectrum $S_D(f)$ extended over $[f_c - f_m, f_c + f_m]$

$$S_D(f) \propto \left| \frac{1}{\sqrt{1 - (f/f_m)^2}} \right|$$

for $f_c - f_m < f < f_c + f_m$



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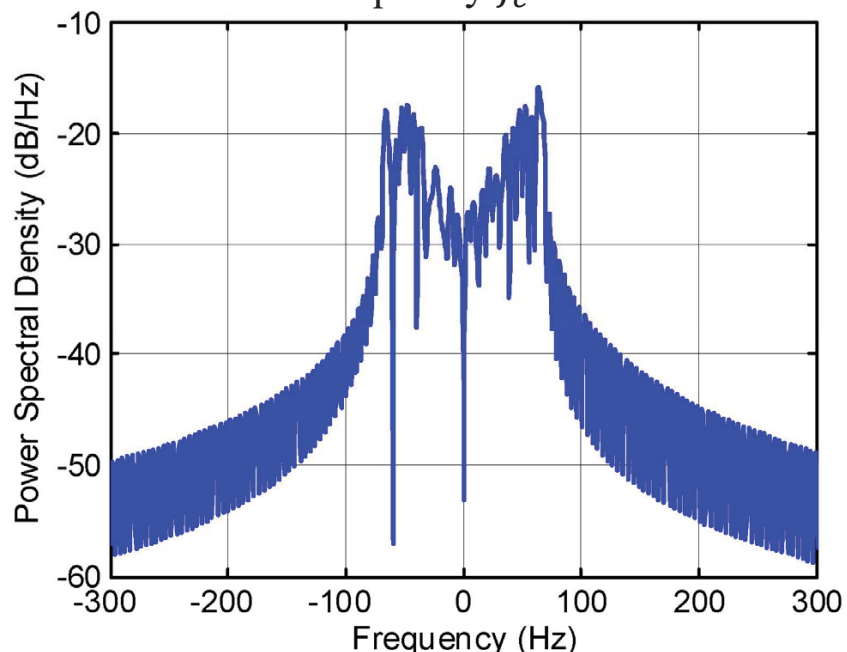
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Doppler Spectrum

□ Jakes model for Doppler spectrum

■ Speed $V=120$ km/h and carrier frequency $f_c=602$ MHz

■ $f_m \approx 67$ Hz



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Coherent Time

- Coherent time T_c is used to characterize the time varying nature of the frequency dispersiveness of the channel in the time domain.
- In general, the Doppler spread and coherent time are inversely proportional to one another.

$$T_c \approx \frac{1}{f_m}$$

- A popular rule of thumb for modern digital communications is, for the coherent time, given as

$$T_c = \frac{0.423}{f_m}$$

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Fast and Slow Fading

- Depending on how fast the channel varying with respect to the symbol period of the Tx. signal, the channel can be divided into as
- Fast fading: $T_{symbol} \gg T_c$
- Slow fading: $T_{symbol} \ll T_c$

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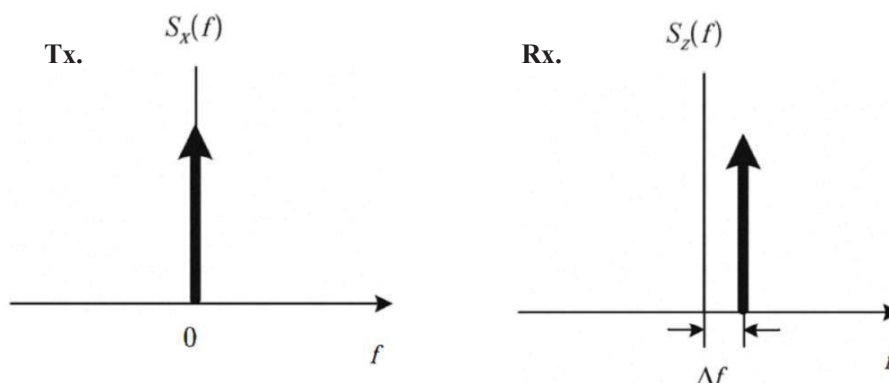
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Additive White Gaussian Noise (AWGN)

- ❑ It is caused by electronic and thermal noise.
 - Exist in analog front-end, digital circuit and other components on PCB
 - Include the flicker and the short noises resulted from the electronic devices
 - Without loss generality, all of above noises are modeled as an additive complex Gaussian noise with zero mean in the baseband equivalent channel model.
 - Both the real and the imaginary parts are $\sigma_v^2/2$ with independent and identically distribution (i.i.d).

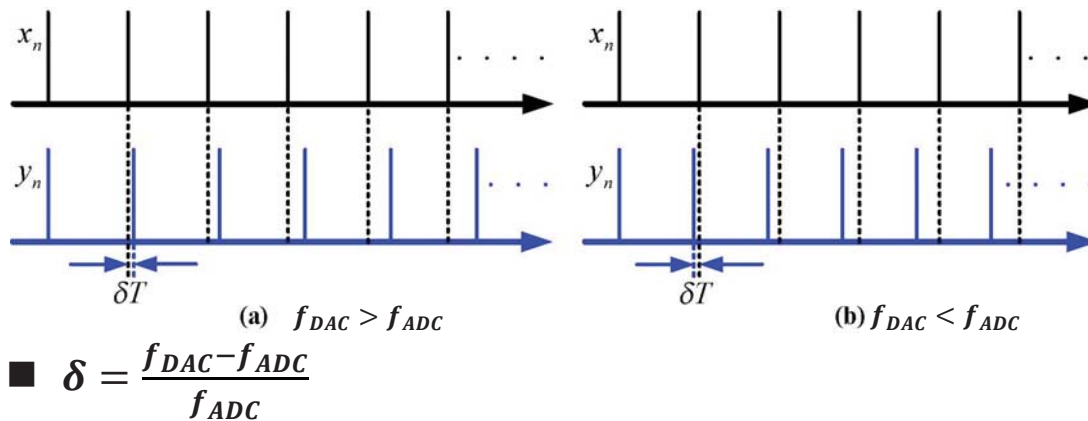
Carrier Frequency Offset (CFO)

- ❑ The local oscillator signal for down-conversion in the Rx. does NOT synchronize with the carrier signal contained in the received signal.
 - Frequency mismatch in the Tx. and Rx.
 - Doppler effect as the Tx. and/or the Rx. in moving



Timing Frequency Offset (TFO)

- ❑ Also called as sampling clock offset (SCO)
- ❑ It is completely the same as CFO since TFO (SCO) is caused by the mismatch of oscillator frequency between transmitter and receiver, and the Doppler effect.



I/Q Imbalance and DC Offset

- ❑ Direct conversion receive structure
 - Zero intermediate frequency (IF) or homodyne receiver
 - Translate the RF (passband) signal from the carrier frequency f_c to DC (baseband)
- Advantages
 - No IF stage
 - No image rejection filter
 - Easy integration due to few components
- Drawback
 - I/Q imbalance
 - DC offset

I/Q Imbalance

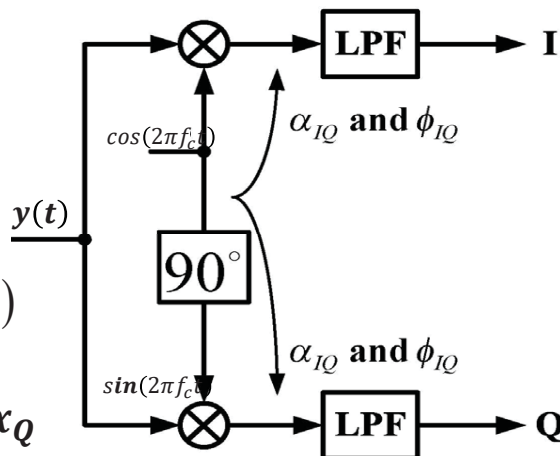
□ Direct conversion

- Use two quadrature mixer to perform the quadrature down-conversion

□ Signal model

$$y(t) = \text{Re}\{x(t)e^{j2\pi f_c t}\} \\ = x_I(t)\cos(2\pi f_c t) - x_Q(t)\sin(2\pi f_c t)$$

- Tx. Baseband signal $x = x_I + j x_Q$
- Gain error α_{IQ} and phase error ϕ_{IQ}



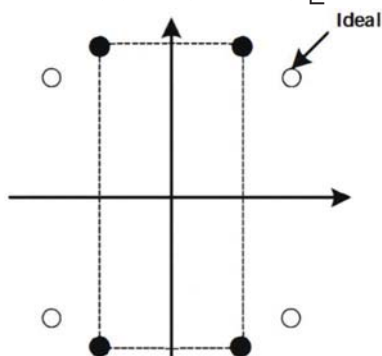
I/Q Imbalance (cont'd)

□ The I/Q imbalance model using mismatch local oscillator

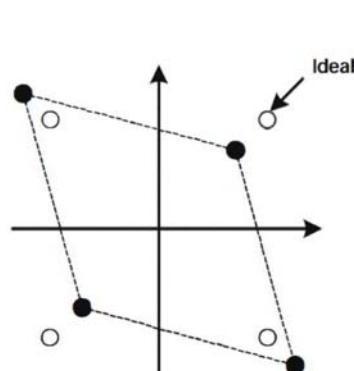
$$2(1 + \varepsilon_A)\cos(2\pi f_c t - \varepsilon_\theta / 2) \\ - 2(1 - \varepsilon_A)\sin(2\pi f_c t + \varepsilon_\theta / 2)$$

□ The demodulated baseband signals with I/Q imbalance are given as

$$\tilde{x}_I(t) = (1 + \varepsilon_A) \left[x_I(t)\cos(\varepsilon_\theta / 2) - x_Q(t)\sin(\varepsilon_\theta / 2) \right] \\ \tilde{x}_Q(t) = (1 + \varepsilon_A) \left[x_Q(t)\cos(\varepsilon_\theta / 2) - x_I(t)\sin(\varepsilon_\theta / 2) \right]$$



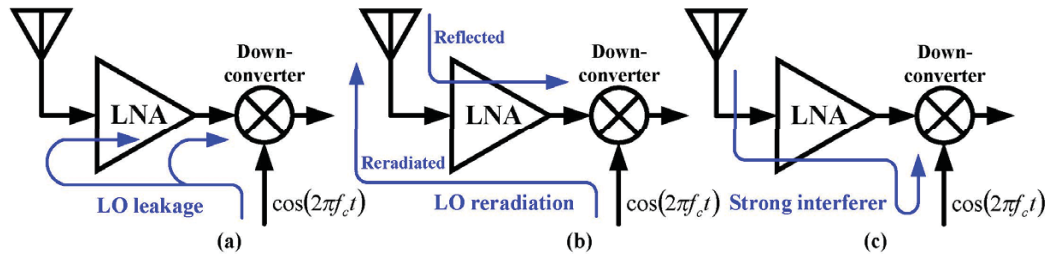
(a) Gain error



(b) Phase error

DC Offset

- It arises from self-mixing and nonlinearity in the front-end.



- (a): Self-mixing caused by the limited isolation of silicon-based ICs between LO and RF ports of mixer
- (b): The LO signal may also leak to the antenna and reradiate. The reradiated LO signal may reflect off obstructions and be recaptured by the front-end, resulting in a time-varying or dynamic DC offset.
- (c): Strong interferer may lead to DC-offset by coupling a path to mixer LO port and then mixing with itself.

3GPP Channel

- Default mobile speed for 3GPP (3rd Generation Partnership Project)

Channel Model	Mobile Speed	Tap	Delay (μ s)	Average Relative Power (dB)	Doppler Spectrum
Typical Urban	3 km/hr	1	0	-5.7	Classical
		2	0.217	-7.6	Classical
		3	0.512	-10.1	Classical
		4	0.514	-10.2	Classical
Rural Area	120 km/hr	5	0.517	-10.2	Classical
		6	0.674	-11.5	Classical
		7	0.882	-13.4	Classical
Hilly Terrain	120 km/hr	8	1.230	-16.3	Classical
		9	1.287	-16.9	Classical
		10	1.311	-17.1	Classical
		11	1.349	-17.4	Classical
Typical urban area (right)		12	1.533	-19.0	Classical
		13	1.535	-19.0	Classical
		14	1.622	-19.8	Classical
		15	1.818	-21.5	Classical
		16	1.836	-21.6	Classical
		17	1.884	-22.1	Classical
		18	1.943	-22.6	Classical
		19	2.048	-23.5	Classical
		20	2.140	-24.3	Classical

3GPP Channel (cont'd)

□ Rural area (left)

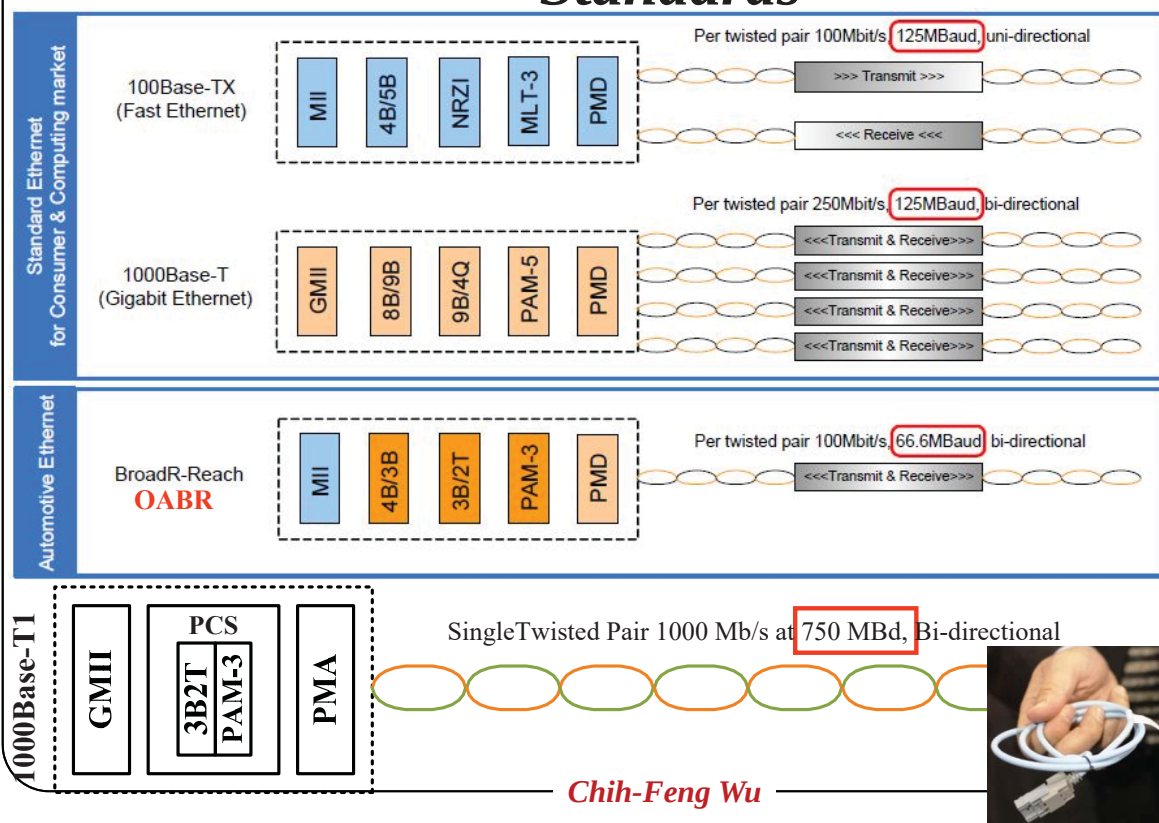
Tap	Delay (μs)	Average Relative Power (dB)	Doppler Spectrum	Tap	Delay (μs)	Average Relative Power (dB)	Doppler Spectrum
1	0	-5.2	Direct path	2	0.356	-8.9	Classical
2	0.042	-6.4	Classical	3	0.441	-10.2	Classical
3	0.101	-8.4	Classical	4	0.528	-11.5	Classical
4	0.129	-9.3	Classical	5	0.546	-11.8	Classical
5	0.149	-10.0	Classical	6	0.609	-12.7	Classical
6	0.245	-13.1	Classical	7	0.625	-13.0	Classical
7	0.312	-15.3	Classical	8	0.842	-16.2	Classical
8	0.410	-18.5	Classical	9	0.916	-17.3	Classical
9	0.469	-20.4	Classical	10	0.941	-17.7	Classical
10	0.528	-22.4	Classical	11	15.000	-17.6	Classical
				12	16.172	-22.7	Classical
				13	16.492	-24.1	Classical
				14	16.876	-25.8	Classical
				15	16.882	-25.8	Classical
				16	16.978	-26.2	Classical
				17	17.615	-29.0	Classical
				18	17.827	-29.9	Classical
				19	17.849	-30.0	Classical
				20	18.016	-30.7	Classical

□ Hilly terrain area (right)

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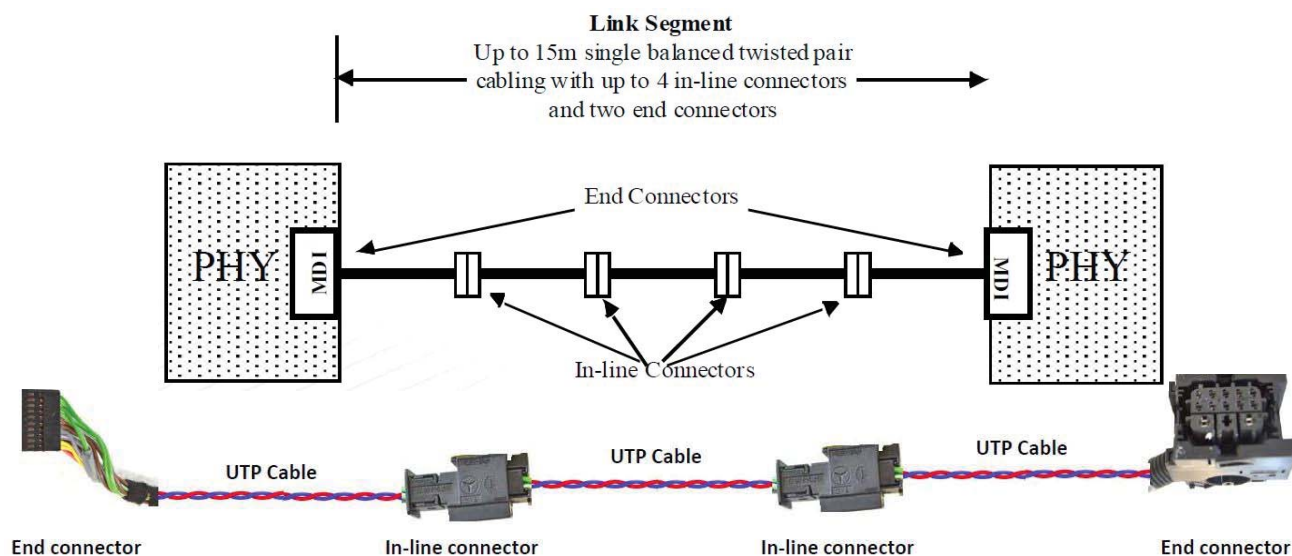
Wireline Communication -- Ethernet PHY Standards



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Channel Characteristics for 1000Base-T1 Connection



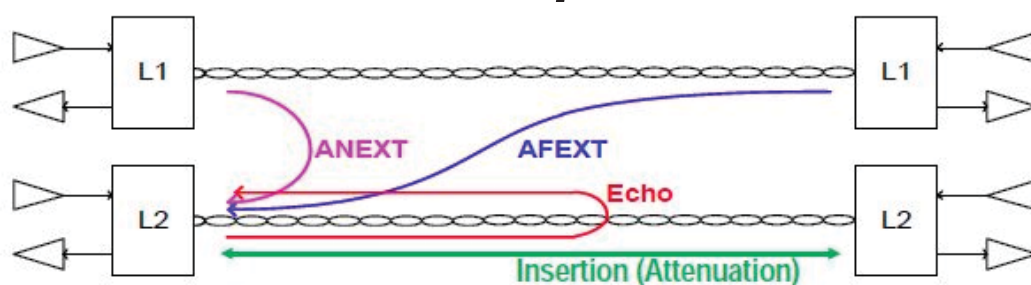
□ End/in-line connectors

- 2-pin or 2-pin of multi-pin connector depending on applications

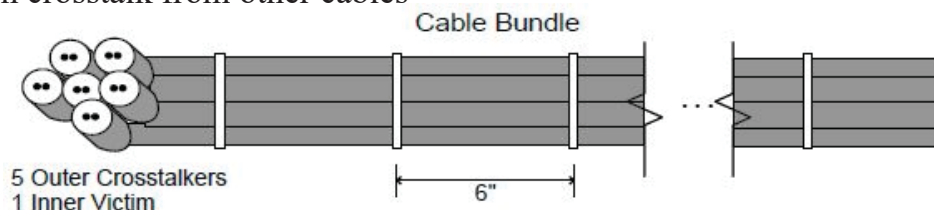
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Channel Impairments



□ Alien crosstalk from other cables

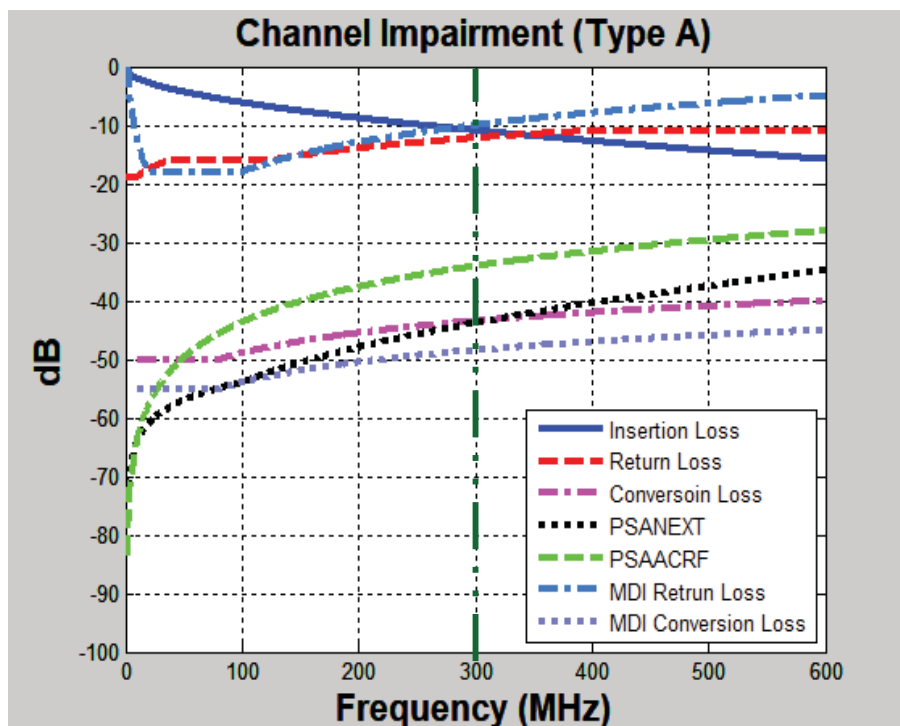


- ANEXT: Alien near-end crosstalk
 - PSANEXT: Power sum Alien NEXT
- FNEXT: Alien far-end crosstalk
 - PSAACRF: Power Sum Attenuation to Alien Crosstalk Ratio Far-End

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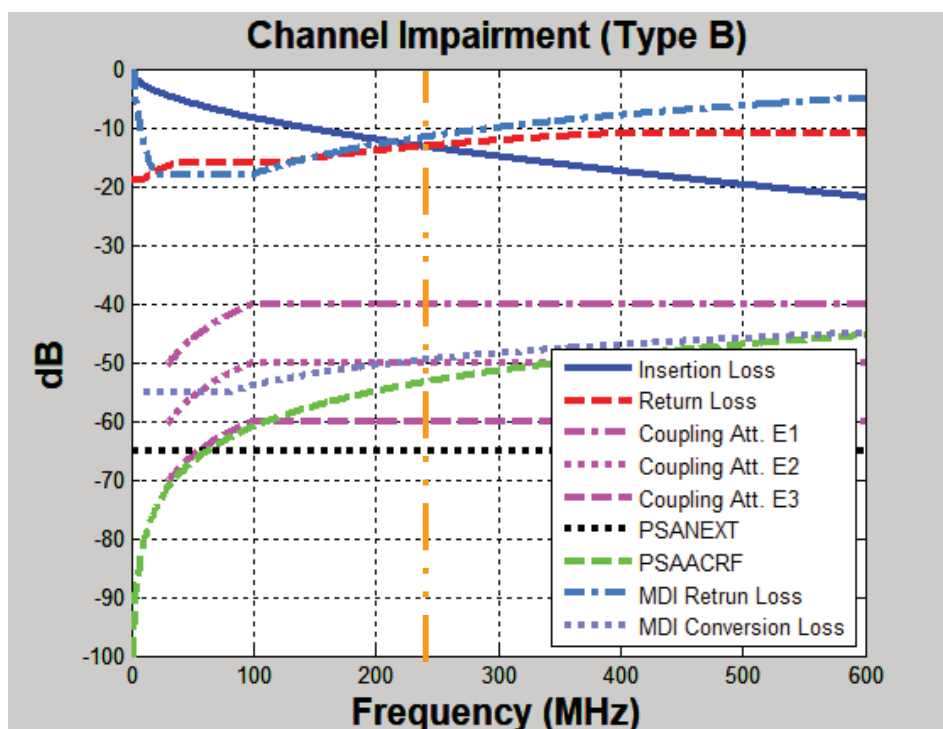
Frequency Response of Channel Impairments



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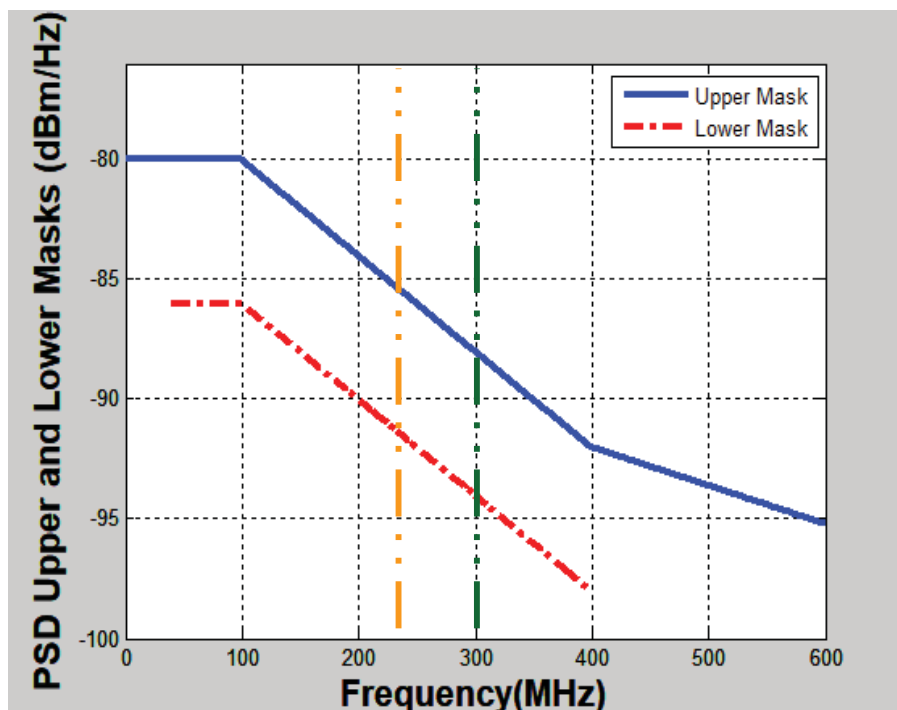
Frequency Response of Channel Impairments



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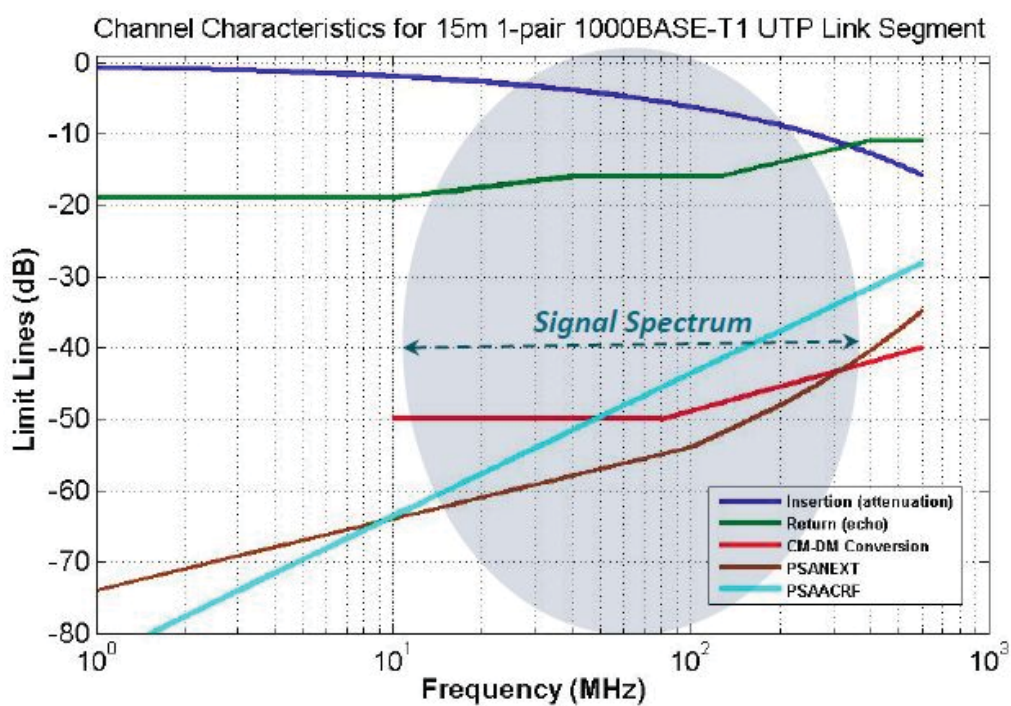
Tx PSD Mask



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Channel Impairments (cont'd)

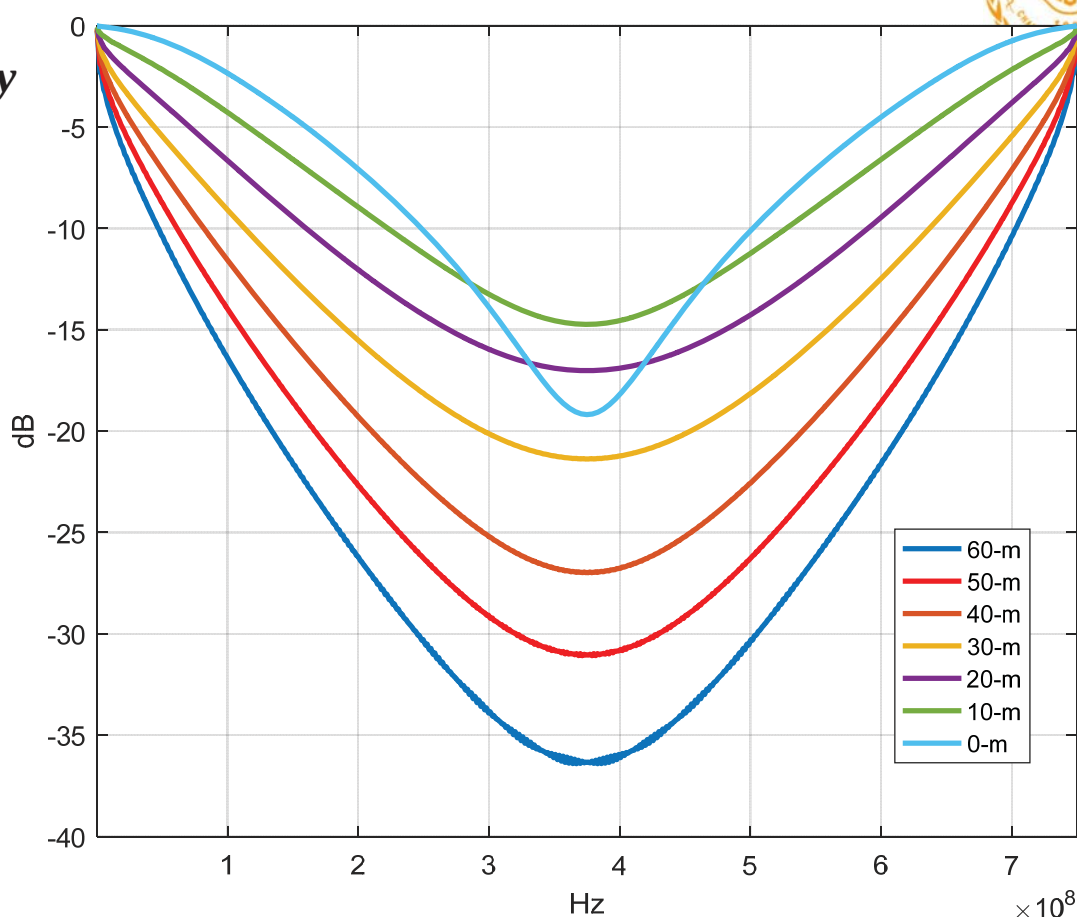


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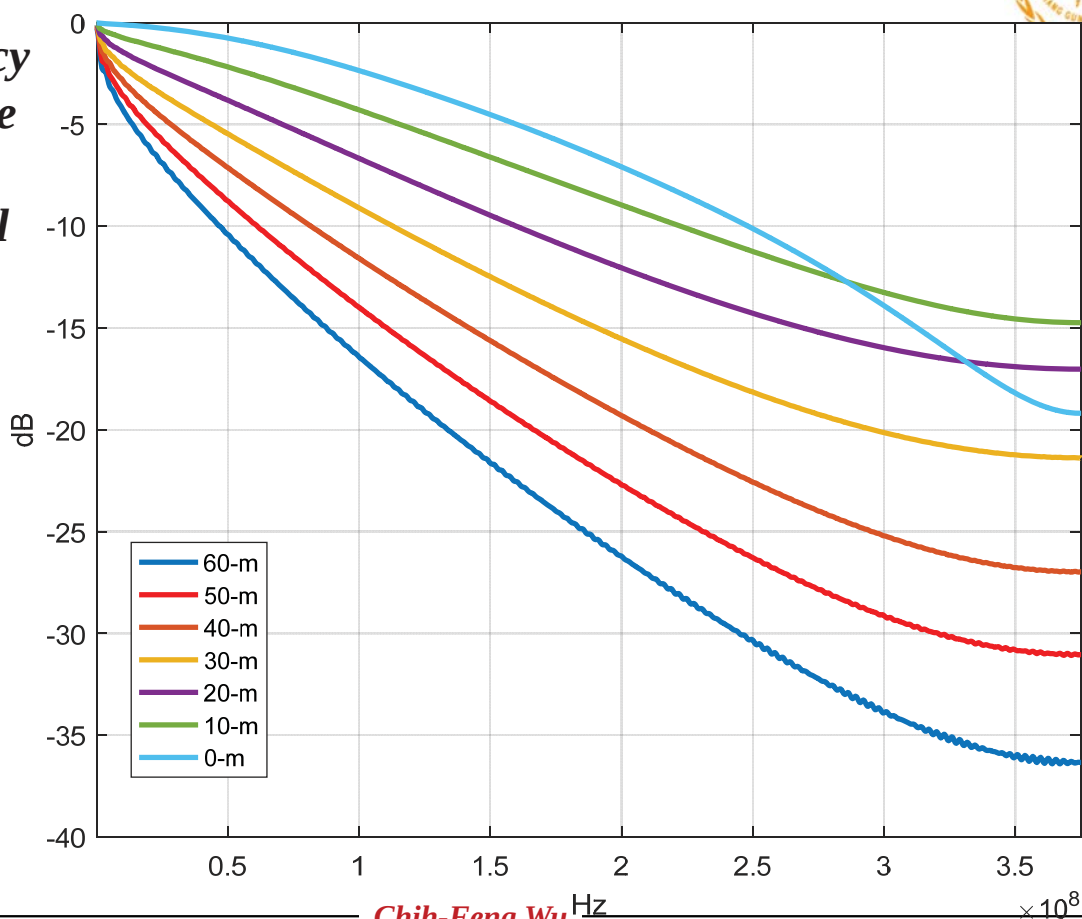
Frequency Response of T1 Channel



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Frequency Response of T1 Channel



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